

Empirical Proof Record: VALIDATED

2026-05-23T23:09:41.663048+00:00

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Proof ID: 98cafc15f11214e26b550eb4a08d86a7...

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1. Claim

Two independent invocations of the 'crdt-laws' scenario with constructor kwargs {'n_sequences': 5, 'ops_per_sequence': 5, 'seed': 20260517} produce bit-identical reproducibility-form proof hashes. (RFC 0002 Phase E4: research-grade reproducibility — every scenario must be deterministic given a fixed seed + corpus + substrate_commit + ophamin_commit.)

- **Operationalisation:** Instantiate the target scenario twice with identical kwargs; run each to emit a signed proof record; compute reproducibility_hash(proof) on each (SHA-256 over the canonical body with wall-clock fields stripped); assert the two hashes match exactly.
- **Threshold:** reproducibility_hash_match >= 1.0 proportion
- **H0:** Two independent runs of the target scenario produce different reproducibility-form hashes — non-determinism is leaking somewhere in the measurement pipeline.
- **H1:** Two independent runs produce bit-identical reproducibility-form hashes — the scenario honours the seed + config contract.

2. Verdict

VALIDATED — observed 1.0

match: reproducibility_hash_a=d235bb996604fb4e..., reproducibility_hash_b=d235bb996604fb4e...
— target scenario was 'crdt-laws'

3. Evidence

Pillar	Statistic	Value	95% CI	Library
reproducibility_audit	reproducibility_hash_match	1.0	—	ophamin 0.115.2

4. Pre-registration

- Registered at: 2026-05-23T23:09:41.457643+00:00
- Config hash: 7ffe47f60a2a3087c17fd9d7cc332ccc...
- Data hash: 7ffe47f60a2a3087c17fd9d7cc332ccc...

5. Signature

0b6d10b704e574bf7882f3522dace63f06498d06d0ba3def9ac0ae4b1f1dc16c